

MedWorld-JEPA: Learning a Medical World Model from Longitudinal Observations

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Abstract

001 *Medical world models aim to anticipate how a patient's*
002 *condition evolves. Realizing this goal requires a predictive*
003 *state that connects clinical meaning with visual evidence.*
004 *We introduce MedWorld-JEPA, a medical world model that*
005 *learns the dynamics of multimodal patient states through*
006 *joint-embedding prediction. Images and reports define a*
007 *shared latent state space, and actual follow-ups supervise*
008 *its temporal evolution. A multi-depth encoder constructs*
009 *compact states from complementary vision-language and*
010 *visual slots. Clinical and spatial task supervision first es-*
011 *tablishes the content of these states; a horizon-conditioned*
012 *world model then learns to predict the state encoded from a*
013 *real follow-up image-report pair. This JEPA objective oper-*
014 *ates directly in the multimodal state space and updates the*
015 *current-state encoder together with the dynamics predictor.*
016 *Future observations enter only the target branch. The result-*
017 *ing model provides two inference pathways: predicted states*
018 *support future finding and report readouts, while current*
019 *states support clinical understanding and image-conditioned*
020 *spatial tasks. Our evaluation protocol examines clinical*
021 *status, progression, report fidelity, and probabilistic reli-*
022 *ability in forecasting, alongside six downstream tasks for*
023 *current-state reuse. [TBD: add the principal forecasting and*
024 *downstream results from Tables 1 and 2].*

025 1. Introduction

026 World models learn internal states and their dynamics to an-
027 ticipate how an observed environment may evolve. In visual
028 learning, this approach connects perception with prediction
029 and planning [2, 27]. Medicine presents a compelling set-
030 ting for such models: repeated examinations record a patient
031 whose findings may persist, resolve, or progress. A medical
032 world model should encode the evidence available at one
033 examination and predict the state that a later examination
034 may reveal. This requires learning both what constitutes a
035 medically informative state and how that state evolves across

observations.

The choice of state is central to this problem. Medical im-
ages provide spatial evidence, while reports express findings
and their clinical interpretation. Follow-up studies capture
changes in both, alongside variation in acquisition and de-
scription. Reproducing a future image or report does not
by itself specify which aspects of the patient the dynamics
model should represent. A useful predictive space must
connect clinical meaning with visual evidence and make
observations from different time points comparable. The
challenge is to construct a latent medical world whose states
have clinical and spatial content, and whose transitions can
be learned from real patient follow-ups.

Joint-embedding predictive architectures (JEPAs) offer a
learning principle for this world. By predicting target em-
beddings, I-JEPA and V-JEPA learn structure directly in rep-
resentation space [1, 6]. VLA-JEPA further couples VLM-
derived tokens with a latent world model for future visual-
state prediction [18]. Medical world modeling has also be-
gun to address patient evolution: MeWM uses generative
tumor simulation, CLARITY learns treatment-conditioned
latent trajectories, and Clin-JEPA combines EHR dynamics
with reusable risk representations [10, 22, 23]. These ad-
vances motivate a complementary direction: a JEPA-based
medical world model whose predictive state is grounded
jointly in images and clinical language, with clinical and spa-
tial tasks shaping the content on which its dynamics operate.

We introduce MedWorld-JEPA, which couples a multi-
modal state encoder with horizon-conditioned dynamics in
a shared latent space. The current image and available re-
port are encoded into state tokens; the world model predicts
their evolution toward the state encoded from an observed
follow-up image-report pair. This gives JEPA a medical
prediction target that combines visual evidence with clinical
semantics. To construct that target space, learned readouts
aggregate multiple depths of vision-language fusion and the
VLM's native vision encoder into four fusion slots and four
visual slots. State construction precedes both the task request
and the forecast horizon, so the same medical observation
defines the starting point for different readouts and temporal

predictions.

Learning this world requires supervision for both state content and state evolution. We first train the state encoder and decoders with clinical and spatial tasks, giving the latent space content that can be read out for medical image understanding. Longitudinal training then teaches a predictor to map the current state and a requested horizon to the encoded follow-up. The prediction loss also updates the current-state encoding path, allowing the demands of temporal modeling to shape the states themselves. Future images and reports supply detached state targets and supervised forecast losses; they are unavailable as inference evidence. The two stages thus connect task-supervised state construction with JEPA-style learning of observational patient dynamics.

This world model provides two forms of inference. Its predicted future state augments a native VLM decoder that retains the current image and available clinical history when predicting follow-up findings and reports. Its current state supports classification, VQA, report generation, phrase grounding, segmentation, and super-resolution. For the dense tasks, visual slots condition decoders that also receive the input image, preserving direct access to spatial detail. We assess forecasting across clinical status, disease progression, report fidelity, and probabilistic reliability (Tab. 1), and current-state reuse through the six downstream tasks (Tab. 2). The downstream protocol withholds the same-exam report and plans matched task supervision and decoder budgets for a no-slots VLM baseline. The tables report available public-model baselines and forecast-pilot results; evaluation of the complete proposed architecture remains pending.

Our contributions are:

- A JEPA-based medical world model that learns horizon-conditioned patient-state dynamics by predicting the joint representation of real follow-up images and reports.
- A multi-depth multimodal world state with complementary vision–language and visual slots, shaped by clinical and spatial task supervision.
- A two-stage learning procedure that establishes the world state through task readout and then adapts its encoding through longitudinal prediction, connecting future forecasting with current-state reuse.

2. Related Work

2.1. Clinical Multimodal Representation

Medical image–text pretraining uses paired images and reports to learn clinical semantics. BioViL demonstrates the value of radiology-specific language modeling for cross-modal alignment and transfer [7]. Medical VLMs extend this foundation to instruction following and generation: LLaVA-Med learns biomedical conversations from figure–caption pairs and synthetic instructions, while MAIRA-2 connects report statements to image regions [5, 16]. M3D

and Med3DVLM extend multimodal analysis to volumetric imaging [3, 21].

These capabilities motivate using a VLM’s internal representations as an interface to other predictors. LISA provides a concrete example: the hidden embedding of a segmentation token conditions a mask decoder alongside visual features [15]. Its token represents a requested segmentation output. Our state is constructed before the task request, with learned readouts of both fused vision–language features and native visual features. Clinical task supervision shapes this shared interface, and spatial decoders combine its visual slots with image evidence. The same state space then defines the targets for longitudinal prediction, connecting multimodal understanding to a reusable predictive representation.

2.2. Predictive Representation Learning

Joint-embedding predictive architectures learn representations by predicting latent targets. I-JEPA predicts embeddings of masked image regions, and V-JEPA extends feature prediction to video [1, 6]. VL-JEPA applies the principle across modalities by predicting continuous text embeddings for vision–language tasks [9]. VLA-JEPA is a direct architectural precedent: VLM-derived latent action tokens condition a world model that predicts future visual states, with future frames confined to the target pathway [18]. Its tokens encode transition information for robotic control, while the predicted states come from a visual encoder. In MedWorld-JEPA, the predicted object is the multimodal observation state itself, combining fusion and visual slots. We first establish this state through clinical task supervision, then learn transitions from actual follow-ups; the conditioning variable specifies an observation horizon.

Within medical imaging, CheXWorld learns radiographic representations through prediction of local anatomy, global layout, and appearance-domain variations [26]. X-WIN learns latent prediction across CT-derived X-ray projections to transfer volumetric anatomical knowledge to radiographs [25]. Their predictive targets capture spatial structure and acquisition transformations. Our temporal targets instead come from repeated observations of the same patient, requiring a representation of clinical change across examinations.

2.3. Medical World Models

Longitudinal medical learning spans interpretation of observed change and prediction of future disease. BioViL-T and TempA-VLP use repeated examinations to learn temporal image–text representations [4, 24], while HERGen incorporates imaging history into report generation [20]. Generative approaches extend this direction to future images: EHRXDiff conditions chest X-ray synthesis on earlier imaging and subsequent medical events, and MedDream simulates disease trajectories through image and video gen-

eration [8, 14].

Medical world models connect disease dynamics to clinical decisions. MeWM combines vision–language treatment planning with generative tumor simulation [22]; CLARITY models treatment-conditioned trajectories in latent space with temporal and clinical context [10]. Clin-JEPA also couples latent EHR trajectory prediction with a reusable backbone for downstream risk tasks [23]. These works establish precedents for both latent patient dynamics and predictive representation learning. Our focus is a shared state of imaging observations and available reports that supports horizon-conditioned follow-up prediction and current clinical and spatial tasks. We therefore assess future prediction and current-state reuse separately, with image evidence retained for dense outputs. This formulation targets observational follow-ups rather than treatment-policy optimization.

3. MedWorld-JEPA

3.1. Task Definition

We study medical state representation and future state prediction from longitudinal observations. At time τ , an observation is $O_\tau = (X_\tau, C_\tau; m_\tau)$, where X_τ is a medical image, C_τ is its report or other admissible clinical text, and $m_\tau \in \{0, 1\}^2$ records modality availability. We consider image-bearing observations, with clinical text when available. Our primary setting uses an image–report pair at both the current time t and an observed follow-up $t^+ > t$.

An encoder E_η maps each observation to multimodal state tokens $S_\tau \in \mathbb{R}^{K \times d}$. Tasks requiring spatial detail additionally receive the input image X_τ directly. For report forecasting, the native vision–language decoder retains the current image and admissible text, augmented with the predicted state. Given S_t and a requested temporal horizon h , a latent world model predicts the follow-up state:

$$\begin{aligned} O_t &\xrightarrow{E_\eta} S_t, \\ (S_t, h) &\xrightarrow{P_\psi} \hat{S}_{t,h} \approx S_{t^+}^*. \end{aligned} \quad (1)$$

The target $S_{t^+}^*$ is encoded from the actual follow-up. We distinguish the requested horizon h from the realized follow-up time t^+ ; their association is set by the temporal sampling protocol. The retrospective forecasting protocol assumes the current report is available at the source observation; other clinical records must satisfy the source-time cutoff. Follow-up images and text supply detached state targets, supervised report/finding targets, and scoring references; they are unavailable as evidence to the predictor or forecast decoder.

The model has two objectives: *predict future states* at a specified horizon and *learn reusable states* for classification, visual question answering, report generation, phrase grounding, segmentation, and super-resolution. State construction precedes both the task request and the horizon condition. A

state represents the available medical observations, without assuming access to all variables governing a patient’s course. Prediction concerns observational follow-ups and carries no treatment-intervention semantics.

Overview. MedWorld-JEPA couples a multimodal state encoder with a horizon-conditioned latent world model (Fig. 1). At both time points, four vision–language slots and four visual slots form the state. Task decoders jointly supervise its construction, with direct image input for segmentation and super-resolution. The world model predicts future slots from the current state and horizon while further adapting the state encoder. For forecasting, these predicted slots augment a native Qwen vision–language decoder that also reads the current radiograph, available report/history, and horizon. This revised architecture remains pending training and evaluation; the completed Table 1 adaptations are legacy prototypes with a different readout.

3.2. State Representation

Shared multimodal state encoding. Current and follow-up observations share an encoding design with $K = 8$ slots of width $d = 1024$. We use image–report pairs for longitudinal prediction; image-based downstream evaluation withholds the same-exam report. State construction precedes both the task query and horizon. The two slot groups retain distinct feature sources throughout their readout and alignment.

Visual and vision–language feature sources. A frozen JEPA encoder f_θ provides spatial features [1, 6, 17], mapped by a trainable visual adapter a_ρ into the VLM embedding space of width d_L :

$$\begin{aligned} D_\tau &= f_\theta(X_\tau) \in \mathbb{R}^{N_D \times d_D}, \\ V_\tau &= a_\rho(D_\tau) \in \mathbb{R}^{N_V \times d_L}. \end{aligned} \quad (2)$$

The VLM fuses V_τ with available text embeddings, yielding layerwise hidden states $H_\tau^{(\ell)}$. In parallel, the VLM’s own vision encoder processes X_τ to produce visual features $F_\tau^{(k)}$ before language fusion. These native visual features supply the last four slots; the JEPA-to-VLM adapter and the slot readouts below have separate roles.

Multi-scale slot construction. As shown in Figure 2, we select four depths from each source: one early layer, two intermediate layers, and the final layer. Let ℓ_j and k_j , $j = 1, \dots, 4$, index the selected fusion and visual layers. Multi-scale refers to this hierarchy of representation depths. The lightweight adapter aggregates each selected feature sequence and projects it to one d -dimensional slot using

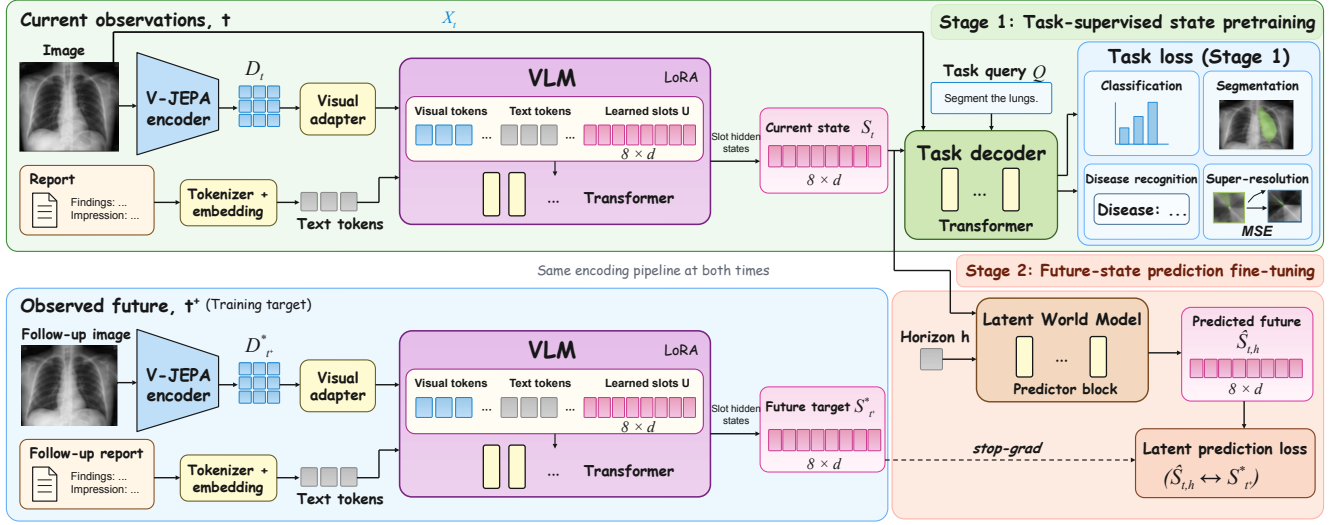


Figure 1. Overview of MedWorld-JEPA. The state $S \in \mathbb{R}^{8 \times d}$ supports task decoding and horizon-conditioned prediction; Figure 2 details its construction. Spatial tasks receive the input image and visual slots, using low-resolution evidence for super-resolution. The world model maps (S_t, h) to $\hat{S}_{t,h}$, supervised by the stop-gradient follow-up target S_{t+h}^* . Stage 1 learns task-supervised states; Stage 2 learns their temporal evolution.

270 source-specific learned readouts r_j^f and r_j^v :

$$\begin{aligned}
 271 \quad s_{\tau,j}^f &= r_j^f(H_\tau^{\ell_j}) \in \mathbb{R}^d, \\
 s_{\tau,j}^v &= r_j^v(F_\tau^{(k_j)}) \in \mathbb{R}^d, \\
 S_\tau &= E_\eta(O_\tau) = [S_\tau^f; S_\tau^v] \in \mathbb{R}^{8 \times d}.
 \end{aligned} \quad (3)$$

272 Each group stacks its four slots in depth order: $S_\tau^f = S_{\tau,1:4}$
 273 and $S_\tau^v = S_{\tau,5:8}$. Alignment gives them a common width
 274 while preserving their sources; visual readouts do not receive
 275 fused or report features. Slots vary with the observation, with
 276 no fixed assignment to diseases, organs, or image regions.
 277 All eight enter the world model without further pooling. The
 278 encoder parameters η comprise the adapters, slot readouts,
 279 and selected VLM parameters.

280 **Task-conditioned readout.** Each task uses a decoder with
 281 parameters ω_q and, when needed, a query Q_q . Classification
 282 and instruction-based disease recognition read all eight slots;
 283 the latter also receives a question and generates a disease list.
 284 Segmentation reads the input image together with only the
 285 four visual slots:

$$286 \quad \hat{M}_\tau = H_{\omega_{\text{seg}}}(X_\tau, S_\tau^v, Q_{\text{seg}}). \quad (4)$$

287 Image-guided feature upsampling motivates this combina-
 288 tion: FeatUp’s guided variant [11] lifts coarse backbone
 289 features to higher spatial resolution using image detail and
 290 multi-view consistency, while UPLIFT [19] combines im-
 291 age guidance with local feature aggregation and multi-scale
 292 reconstruction supervision. We use this complementarity

293 through hidden-state conditioning: the image branch retains
 294 spatially resolved local evidence, and S_τ^v supplies visual con-
 295 text from multiple depths as the decoder constructs dense
 296 outputs. Task losses learn their interaction without impos-
 297 ing a one-to-one slot-to-pixel correspondence or a separate
 298 feature-matching objective.

299 Super-resolution likewise reads image pixels and four
 300 visual slots, both derived exclusively from the same low-
 301 resolution input:

$$\begin{aligned}
 \hat{X}^{\text{HR}} &= H_{\omega_{\text{SR}}}(X^{\text{LR}}, S^{\text{v,LR}}), \\
 \mathcal{L}_{\text{SR}} &= \frac{1}{N} \left\| \hat{X}^{\text{HR}} - X^{\text{HR}} \right\|_F^2.
 \end{aligned} \quad (5)$$

303 Here $S^{\text{v,LR}}$ is obtained by applying the visual encoder and
 304 its four readouts to X^{LR} . The HR image supplies supervi-
 305 sion only. N counts valid target pixels, excluding padding;
 306 prediction and target share an intensity scale. Neither spa-
 307 tial decoder reads S^f . The expanded six-task evaluation
 308 adds standard VQA, report generation, and phrase ground-
 309 ing through the adaptation protocol.

3.3. State Prediction 310

311 **Horizon-conditioned latent dynamics.** The latent world
 312 model takes the current state tokens and an embedding of the
 313 requested horizon, and predicts a state with the same token
 314 shape:

$$315 \quad \hat{S}_{t,h} = P_\psi(S_t, h) \in \mathbb{R}^{K \times d}. \quad (6)$$

316 The predictor receives both slot groups, with no additional
 317 image or feature-map input. Latent prediction on pretrained
 318 features has precedent in DINO-WM [27]; our predictive

space combines fusion and visual slots. Prediction is direct at h ; multi-time-point records supply training pairs without requiring autoregressive rollout.

Observed future states as targets. Let $E_{\bar{\eta}}$ denote the target instance of the same encoding design. The actual follow-up provides the target for latent prediction:

$$\begin{aligned} S_{t+}^* &= E_{\bar{\eta}}(O_{t+}) \in \mathbb{R}^{K \times d}, \\ \mathcal{L}_{\text{future}} &= \ell_S(\hat{S}_{t,h}, \text{sg}(S_{t+}^*)), \end{aligned} \quad (7)$$

where sg stops target-branch gradients. The follow-up image supplies both feature pathways, and its report contributes to the fusion slots. This latent objective supervises the eight future slots, with no separate dense feature-map loss. Future observations and their modality indicators are not predictor inputs.

The fixed source and depth ordering establishes correspondence between predicted and target slots. The latent distance and normalization remain to be selected; element-wise mean squared distance is one candidate. Current spatial decoders also require the corresponding input image and therefore do not directly define dense future outputs.

Native report forecasting with predicted-state context.

Let A_γ project the eight predicted slots to the native Qwen language width. The forecast decoder retains Qwen’s native vision encoder and image–text interface:

$$(\hat{R}_{t,h}, \hat{p}_{t,h}) = D_\omega(X_t, C_t, h; A_\gamma(\hat{S}_{t,h})). \quad (8)$$

Here C_t includes the current report and source-available clinical history. Native visual tokens from X_t , the current text, and the horizon remain in the decoder context; projected future slots are inserted at the end of the native user message, before the assistant generation prompt. The finding head reads the final source-prompt hidden state before any teacher-forced future report tokens. Both native supervised fine-tuning and the state-augmented condition use the same aspect-preserving 512-pixel source image, native decoder/finding-head initialization, and report objective. Encoder and decoder share immutable pretrained weights but have independent LoRA parameters. Follow-up images never enter this context; follow-up report tokens are used only as causally masked training targets.

The native supervised baseline omits $A_\gamma(\hat{S}_{t,h})$. Its comparison with MedWorld-JEPA measures the contribution of the added state branch, which includes a JEPA encoder and learned dynamics. Isolating prediction and compression further requires controls that retain the same encoders and supply current slots or full tokens. In the shuffled control, only the state branch uses a different patient’s source observation from the same split; the native decoder retains the true current image, text, and requested horizon.

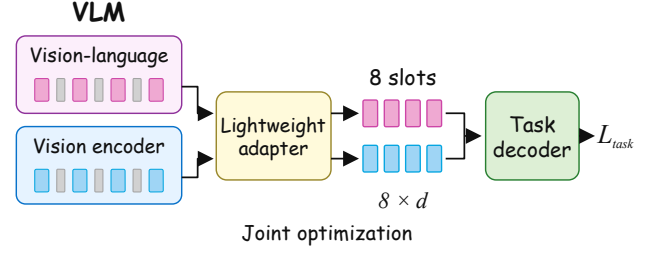


Figure 2. Construction of the eight state slots. Colored blocks select one early, two intermediate, and the final layer from each source. Lightweight readouts form four vision–language and four visual slots; task losses jointly train the readouts and decoders.

3.4. Two-Stage Training

Single-time-point annotations supervise state content; longitudinal pairs supervise its evolution. Related staged VLM/world-model adaptation appears in VLA-JEPA [18]. We first establish a task-supervised medical state space and then learn longitudinal dynamics within it.

Stage 1: task-supervised state pretraining. Starting from pretrained JEPA and VLM weights, we jointly train the adapters, multi-scale slot readouts, task decoders, and selected VLM parameters using parameter-efficient adaptation [12]; JEPA remains frozen. For the tasks $\mathcal{Q}_i$ supported by sample i , we minimize

$$\mathcal{L}_{\text{task}} = \mathbb{E}_i \left[\sum_{q \in \mathcal{Q}_i} \alpha_q \ell_q(\hat{y}_i^{(q)}, y_i^{(q)}) \right], \quad (9)$$

where each prediction uses the task-specific slot subset and inputs defined above. Classification uses binary cross-entropy, report generation and disease recognition use token cross-entropy, segmentation uses Dice and pixel cross-entropy, and SR uses Eq. (5).

There is no stop-gradient between the task decoders and slot readouts. Task gradients flow through S into the learned aggregation/projection modules and any adapted source layers, jointly with decoder updates. Classification and disease recognition supervise both slot groups; segmentation and SR directly supervise S^v and their image decoders. Thus training updates the shared parameters that produce observation-dependent slots, rather than treating each sample’s slots as independent optimization variables.

Stage 2: future state prediction fine-tuning. We initialize the encoder from a newly trained 4 + 4 Stage 1 model and train P_ψ on same-patient current–future observations, primarily MIMIC-CXR image–report pairs [13]. The legacy final-language-layer checkpoint does not initialize this state design. Future latent, report, and finding losses update P_ψ

and the current-state construction modules, while the latter two also train the native forecast decoder; JEPA remains frozen. With optional current-task replay,

$$\mathcal{L}_{\text{stage2}} = \lambda_S \mathcal{L}_{\text{future}} + \lambda_R \mathcal{L}_{\text{report}} + \lambda_F \mathcal{L}_{\text{finding}} + \beta \mathcal{L}_{\text{task}}, \quad (10)$$

where losses are averaged over sampled training pairs, $\mathcal{L}_{\text{report}}$ is future-report token cross-entropy, and $\mathcal{L}_{\text{finding}}$ is masked binary cross-entropy on future findings. Setting $\beta = 0$ omits current-task replay. The native supervised baseline uses the same report and finding losses, without a latent target. Temporal prediction and reusable representation are evaluated separately on held-out patients.

Target updates and implementation choices. The target encoder is a frozen snapshot of the new 4 + 4 Stage 1 encoder; its parameters remain fixed throughout Stage 2. Its immutable pretrained weights can share storage with the online branches, while its copied adapters and readouts remain separate and receive no updates. Stop-gradient excludes the target pass from future-loss backpropagation. The initial implementation uses classification and current-report state supervision together with native current-report decoder training in Stage 1; this subset is not the complete six-task protocol. [TBD: validate the revised native decoder, selected layer/readout configuration, and matched training budgets before reporting new results; see the supplementary material].

4. Results

4.1. Experimental setup

We report partial future-prediction and downstream results in Tabs. 1 and 2. MIMIC-CXR [13] supplies images and reports, with source-available MIMIC-IV history for forecasting. Forecast training/validation/test sets contain 16,000/230/297 pairs; the test set contains 94 patients. Adapted representation models use 2,400 updates with batch size 32. The three legacy Qwen3.5-9B forecast conditions complete the same budget from a shared 9B Stage 1 initialization. Dense tasks use 4,096/249/447 images and 20 training epochs; human lung Dice is evaluated separately on 138 Montgomery images. The tables distinguish zero-shot checkpoints, frozen-feature task heads, and trained forecast adaptations. The completed 9B slot adaptation uses final-layer language queries; it does not implement the proposed four fusion-layer plus four native-vision slots. New native supervised and predicted-state-augmented Qwen forecasters retain the same 512-pixel current-image and report/history context; their training and results are pending. Their planned protocol uses a new 1,694-update Stage 1 at effective batch eight, followed by 2,400 forecast updates at effective batch 32, subject to 0.8B and 9B execution and memory checks.

Full six-task adaptation of that design, standard MIMIC-CXR-VQA, and MS-CXR phrase grounding remain pending. All reported numbers are point estimates; matched data and update counts do not imply matched GPU hours.

4.2. Future prediction

Table 1 evaluates four dimensions of the predicted future state: clinical status, disease progression, report fidelity, and probabilistic reliability, with two metrics per dimension. Inputs comprise the current radiograph and report, clinical history available by the source cutoff, and a horizon bin. The current report is assumed available retrospectively; follow-up images and reports supply targets only. Four public VLM checkpoints use zero-shot forecast prompts, with Qwen3.5-9B used for the Qwen reference. BioViL-T and CheXWorld receive trained forecast modules; Copy Current and a finding-transition prior provide simple references. The bottom three rows report legacy Qwen3.5-9B full-token, eight-slot, and cross-patient shuffled-state forecasters. These conditions complete matched forecast training from a shared 9B Stage 1 initialization and evaluation on all 297 test pairs, including GREEN. The slot adaptation uses final-language-layer queries, separate from the proposed multi-depth 4 + 4-slot model. Their decoder reads the predicted representation without the native current-image context. New native supervised and native decoder plus predicted-state conditions remain pending; the legacy full-token row is not native Qwen image-text fine-tuning.

Clinical status uses finding macro AP (non-interpolated average precision) and AUROC. Disease progression uses Transition F1, averaged over supported finding-specific onset and resolution events, and Direction F1, which measures improvement, stability, and worsening of persistent findings. Stable references remain eligible for false-change penalties. RadGraph partial F1 and GREEN assess the clinical content of generated future reports. Probabilistic reliability uses macro Brier score and classwise ECE on the same future-finding probabilities. Zero-shot VLM scores are derived from fixed Yes/No answer likelihoods; trained forecasters use supervised finding heads. The planned native supervised and state-augmented comparison will use matched finding heads. All methods share the reference-defined eligibility masks for each metric; training and readout differences and calibration rules are specified in the supplementary material. Direction annotations remain unadjudicated for this cohort, so that column is pending; the separate 82-pair direction study is not included.

All scores below and in the two tables are multiplied by 100, except PSNR (dB). Qwen3.5-9B reaches AP 85.55 and AUROC 81.77 without task adaptation; CheXagent reaches 86.89/81.48. BioViL-T reaches 84.30/81.24 with the shared trained predictor. Relative to the full-token forecaster, the 9B slots improve Transition F1 (28.51 versus 16.03) and

Table 1. Available future-state results on 297 held-out pairs (94 patients). All scores are multiplied by 100 and shown to two decimals; arrows indicate the preferred direction. ZS denotes public-checkpoint zero-shot inference.

Method	Future clinical status		Disease progression		Future report fidelity		Probabilistic reliability	
	AP ↑	AUROC ↑	Trans. F1↑	Dir. F1↑	RadGraph F1↑	GREEN ↑	Brier ↓	ECE ↓
Qwen3.5-9B ^{ZS}	85.55	81.77	13.60	TBD	20.44	31.70	20.22	27.90
LLaVA-v1.6-Mistral-7B ^{ZS}	82.90	73.96	13.67	TBD	19.24	30.78	21.87	29.28
LLaVA-Med-v1.5-7B ^{ZS}	81.59	73.31	1.92	TBD	18.23	21.24	17.61	17.90
CheXagent-8B ^{ZS}	86.89	81.48	0.00	TBD	13.43	15.51	13.00	13.22
BioViL-T + predictor	84.30	81.24	22.65	TBD	14.70	15.37	10.75	7.73
CheXWorld + predictor	83.80	79.56	23.71	TBD	13.15	12.63	11.73	9.04
Copy Current	—	—	0.00	TBD	20.77	24.58	—	—
Finding-transition prior	75.48	64.60	TBD	—	—	—	13.22	7.19
Full-token forecaster (Qwen3.5-9B)	83.09	76.42	16.03	TBD	17.11	17.83	11.64	9.42
MedWorld-JEPA (Qwen3.5-9B slots)	82.70	77.94	28.51	TBD	15.96	17.61	11.47	8.93
Qwen3.5-9B (shuffled state)	70.93	52.85	6.04	TBD	10.78	8.87	13.33	4.97

AP is non-interpolated average precision; AP/AUROC/Brier/ECE are macro-averaged over findings with common reference masks. RadGraph uses partial F1. VLM scores use Yes/No likelihoods. Trained forecasters use 16,000 pairs, 2,400 updates and batch 32; actual GPU hours differ. “+ predictor” is an official frozen-encoder adaptation with shared forecast modules, not an original published forecasting system. The bottom three rows are legacy JEPA-path Qwen3.5-9B forecasters with shared Stage-1 initialization. Their slots use eight final-language-layer queries, distinct from the proposed multi-depth fusion/vision architecture; the full-token control retains the same predictor. Native-image-conditioned 4+4 results, Direction labels and the prior’s transition readout are pending. The prior reads structured source findings. TBD: pending; —: no score for this interface. Single-seed point estimates.

Table 2. Available results and remaining evaluations on six downstream tasks. Higher is better. All scores are multiplied by 100 except PSNR (dB); values are shown to two decimals. ZS denotes no benchmark-specific adaptation.

Method	Classification		VQA		Report gen.		Grounding		Segmentation		SR 4×	
	AUC	AP	Acc.	μ F1	RG	CB	mIoU	A@.5	Dice pseudo	Dice human	PSNR	SSIM
Qwen3.5-9B ^{ZS}	71.54	85.46	TBD	TBD	23.98	37.97	TBD	TBD	—	—	—	—
LLaVA-v1.6-Mistral-7B ^{ZS}	54.22	72.64	TBD	TBD	6.87	2.43	TBD	TBD	—	—	—	—
LLaVA-Med-v1.5-7B ^{ZS}	54.70	72.96	TBD	TBD	6.81	15.66	TBD	TBD	—	—	—	—
CheXagent-8B ^{ZS}	79.08	87.92	TBD	TBD	16.93	30.60	TBD	TBD	—	—	—	—
DINOv2 + heads	68.78	84.13	—	—	—	—	—	—	93.03	75.54	32.06	90.86
CheXWorld + heads	75.05	86.18	—	—	—	—	—	—	93.08	75.72	32.06	90.87
MAIRA-2	—	—	—	—	TBD	TBD	TBD	TBD	—	—	—	—
SwinIR (CXR-adapted)	—	—	—	—	—	—	—	—	—	—	37.27	94.51
Image-only + heads	—	—	—	—	—	—	—	—	91.20	75.71	31.87	90.81
Qwen3.5-9B vision + slots [†]	—	—	—	—	—	—	—	—	92.31	75.38	31.87	90.82
Qwen3.5-9B vision + shuffled [†]	—	—	—	—	—	—	—	—	91.18	75.66	31.86	90.79
Qwen3.5-0.8B no-slots	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
MedWorld-JEPA (full model)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Classification/report tests contain 353/507 images; classification heads use 13,681 training images and 20 epochs. All filled segmentation/SR rows use 4,096/249/447 train/validation/test images, 20 epochs and batch 8. Human Dice uses 138 external Montgomery lung masks; pseudo Dice uses CXAS. [†]Only the frozen native vision tower is used, with four fixed depth-pooled tokens and trained heads; the 9B language model is not used in these rows. Shuffled tokens come from other patients. SwinIR is fully adapted; its training/evaluation takes 8.31 GPU-h versus 0.16 for image-only SR on the same GPU type, so this is not an equal-GPU-hour comparison. RG/CB: RadGraph/CheXbert F1; A@.5: box accuracy at IoU ≥ 0.5 . TBD: pending; —: not evaluated with this interface. Official VQA, MS-CXR grounding and the full six-task adaptation remain pending; derived QA, anatomy localization and report-assisted prototypes are excluded.

Longitudinal radiographic state forecasting on MIMIC-CXR

Candidate visualization — pending

Figure 3. Planned patient trajectory: an observed chest-radiograph timeline, reference report excerpts, and predicted reports above one or two finding probability curves. All forecasts start from the same Day 0 observation; follow-up images and reports are references only. Comparisons include Copy Current and a matched forecaster. Curves use actual elapsed times, with separate reference markers for positive, negative, and uncertain or unmentioned findings. Predicted probabilities indicate presence, not severity.

AUROC (77.94 versus 76.42), with lower Brier/ECE, but slightly lower AP, RadGraph, and GREEN. Shuffling reduces AUROC to 52.85 and GREEN to 8.87; all 297 generated reports collapse to one output, with zero CheXbert F1. Its lower ECE (4.97) therefore accompanies poor discrimination and report content. These point estimates support a limited state-condition comparison; they do not validate the proposed multi-depth model or establish the effect of native Qwen fine-tuning. A future comparison against native supervised Qwen will test the added state branch as a whole; current-slot and full-token controls with the same encoders are needed to separate prediction and compression effects. Historical 0.8B ablation scores remain archived separately.

4.3. Downstream tasks

Table 2 covers six tasks without the future predictor: classification, standard VQA, current report generation, phrase grounding, organ segmentation, and super-resolution. This expanded image-based evaluation withholds the same-exam report from state construction. Questions and phrases condition task readout after the state is encoded. The no-slots baseline and MedWorld-JEPA remain pending under matched six-task supervision and decoders. Their earlier report-assisted results and separate two-epoch joint dense-task pilots do not populate those rows. Frozen Qwen3.5-9B visual-slot task heads, their shuffled control, and an image-only decoder are reported separately.

Classification uses 13 MIMIC finding/device labels and reports macro AUROC/AP on 353 test images. DINOv2/CheXWorld classification heads use 13,681 training and 160 validation images for 20 epochs. Current-report scoring uses 507 images. MIMIC-CXR-VQA provides standard questions and official Accuracy/micro F1; Verify/Choose/Query and whole-image/region breakdowns will be reported separately. Current report generation uses MIMIC-CXR with RadGraph partial F1 and CheXbert macro-positive F1. MS-CXR phrase grounding uses mIoU and box accuracy at $\text{IoU} \geq 0.5$ under one fixed conversion and matching protocol.

Segmentation measures Dice for the lungs and heart against CXAS pseudo-labels and, separately, human lung masks from Montgomery. Super-resolution uses synthetic $4\times$ downsampling per spatial dimension, with LR imagery

in both input branches and HR used only as the target. PSNR and SSIM exclude padding. DINOv2/CheXWorld use frozen visual encoders with supervised heads; MAIRA-2 and SwinIR provide reporting/grounding and reconstruction references, respectively. The completed SwinIR adaptation reaches 37.27 dB and SSIM 94.51, but consumes 8.313 GPU hours including evaluation, versus 0.161 for the image-only SR run. This is not an equal-compute comparison. Formal VQA, MS-CXR grounding, and the full six-task model rows remain pending; derived QA and anatomical-region localization scores are excluded.

4.4. Patient trajectory

Figure 3 reserves a longitudinal case study with four to five observations from one test patient. The planned analysis predicts each follow-up independently from the same Day 0 radiograph and report, conditioned on the actual elapsed time. Observed follow-up images and reports provide references only. Predicted reports and finding probabilities will be compared with Copy Current and a matched forecaster at the irregular observation times. This is fixed-origin, multi-horizon forecasting; connecting the predictions does not constitute an autoregressive rollout. Case selection and predictions remain pending.

4.5. Attention visualization

Figure 4 reserves a view of how the four visual slots aggregate patches from the current radiograph. The planned visualization uses the actual softmax weights of the learned visual-slot readout at each selected encoder depth, with a shared color scale. These weights describe current-state construction; the accompanying report text supplies case context, not a finding-specific attention query. Slots have no prescribed anatomical roles, and the maps do not establish future lesion locations or forecasting accuracy. Weight export and verification of the token-to-image mapping remain pending.

5. Discussion and Limitations

Future prediction and reusable representation require separate evidence. Clinical-status, progression, report-fidelity, and probability-reliability scores assess forecasting, while the six downstream tasks assess information accessible

Visual-slot attention across representation depths

Candidate visualization — pending

Figure 4. Planned attention visualization: one current radiograph and a brief report excerpt above a 2×2 grid of actual visual-slot attention overlays (S5–S8). Panels identify the source encoder depths and share a color scale and opacity. Weights describe patch aggregation during current-state encoding, not finding-specific attribution or predicted future lesion locations.

through the current state. The generic output slots describe available observations without guaranteeing complete, interpretable, or disentangled clinical states. The planned downstream adaptation does not by itself establish that longitudinal training improves these representations. Dense-task decoders can solve tasks through their direct image branches. Matched image-only controls are therefore necessary to establish any contribution from S_t ; task accuracy alone does not show that the state retains more useful information.

Visual slots remain defined when the report is withheld; performance with and without clinical text requires evaluation. Predicting $\hat{S}_{t,h}$ does not establish image synthesis, full-report generation, or future dense outputs from decoders that also require the corresponding input image. Medical JEPA retraining and transfer of volumetric knowledge to radiographs remain untested directions.

The model learns observational follow-ups. Follow-up selection, treatment, care setting, scanner changes, and historical report content can correlate with recorded futures. Clinical-record timestamp filtering and target-only future evidence constrain the forecasting inputs without identifying natural history or causal treatment effects. Current reports are assumed available retrospectively because reliable signing times are unavailable. Conclusions depend on the evaluated cohorts, horizons, and annotations. Supplementary retrieval additionally depends on candidate-pool size and eligibility; their coverage must be reported.

VLM adaptation makes training cost consequential. Stop-gradient isolates target outputs from the prediction loss but does not ensure stability or prevent collapse. The target-encoder update policy and any continued task supervision in Stage 2 require specification and validation. Future reports must remain unavailable to the prediction inputs.

6. Conclusion

MedWorld-JEPA is a medical latent world model for future state prediction and reusable multimodal representation. It combines multi-depth vision–language and native visual slots, jointly shapes their readouts and decoders through task supervision, and predicts their horizon-conditioned evolution from longitudinal observations. Current dense-task

decoders receive the input image directly and use the four visual slots as a learned condition. Current and future observations use the same encoding design, with future evidence confined to a stop-gradient target branch. [TBD: insert the final future-prediction and downstream-task results]. The two evaluations assess forecasting and the current state’s use across clinical and spatial tasks.

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